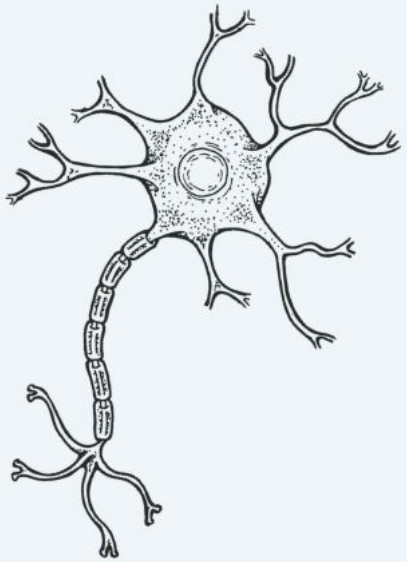
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Predicting Preclinical Neurodegeneration via AI-Decoded Microglial Activation Patterns from Brain Imaging

Sinchana Joshi, Tejshree Ratnakar



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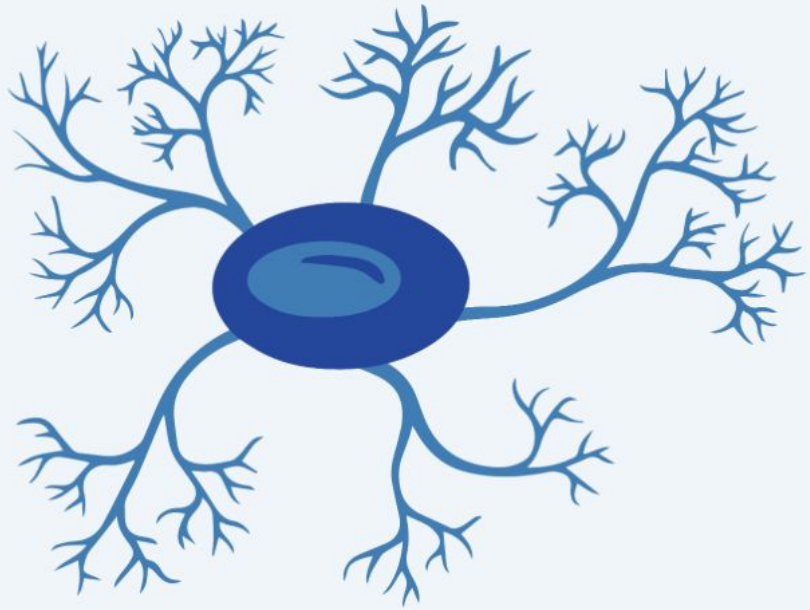
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II Objectives/Research Focus

III Methodology/Device Design

IV Results

V Conclusion & Discussion

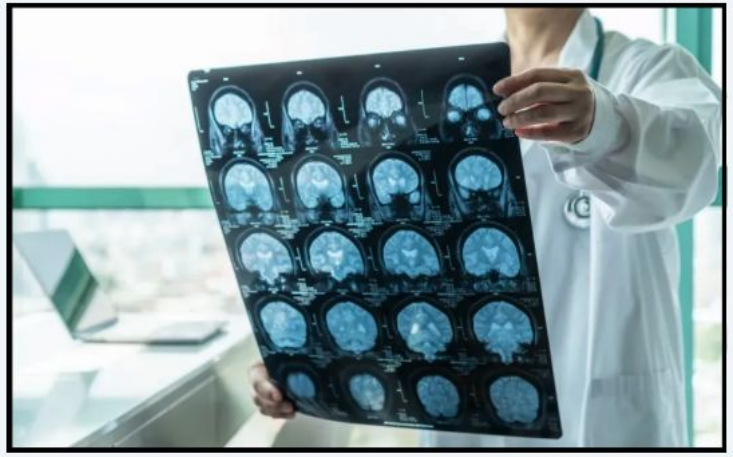


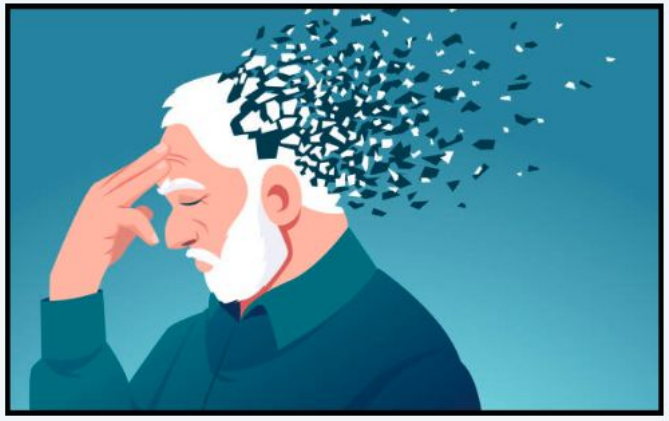
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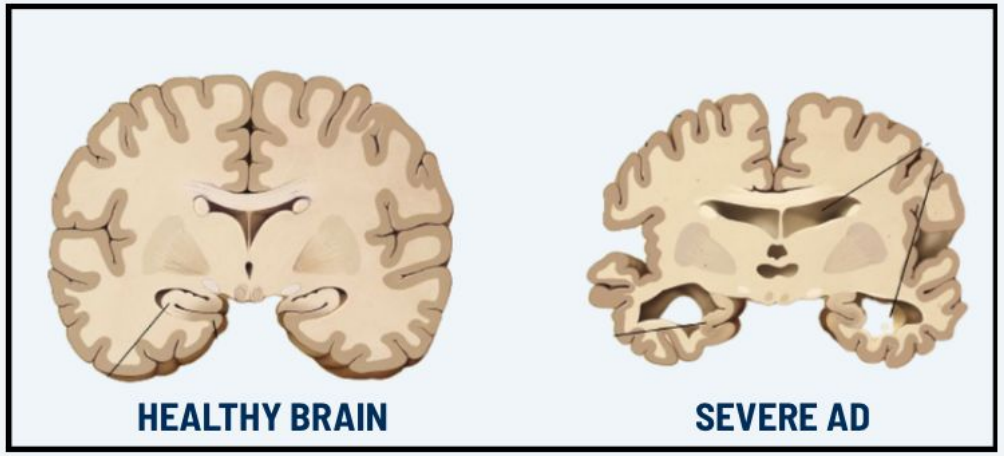
INTRODUCTION

RESEARCH BACKGROUND









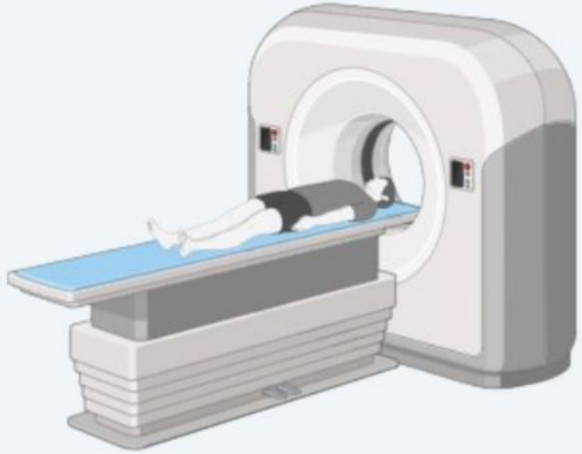
OBJECTIVES/RESEARCH FOCUS

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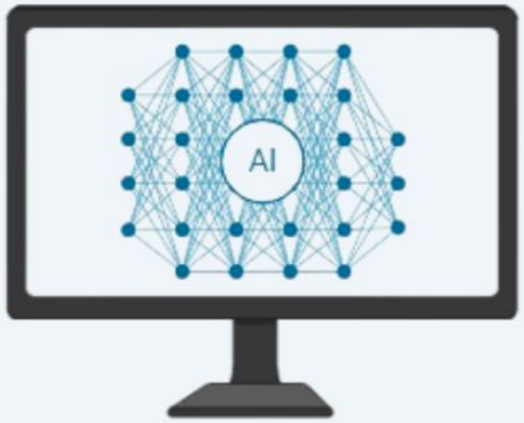
GOAL

Improve accuracy of PET scans in detecting early neurodegeneration through novel technologies



PROCESS

Create an AI model that predicts neurodegeneration through microglia



TESTING

Validate accuracy on synthetic data

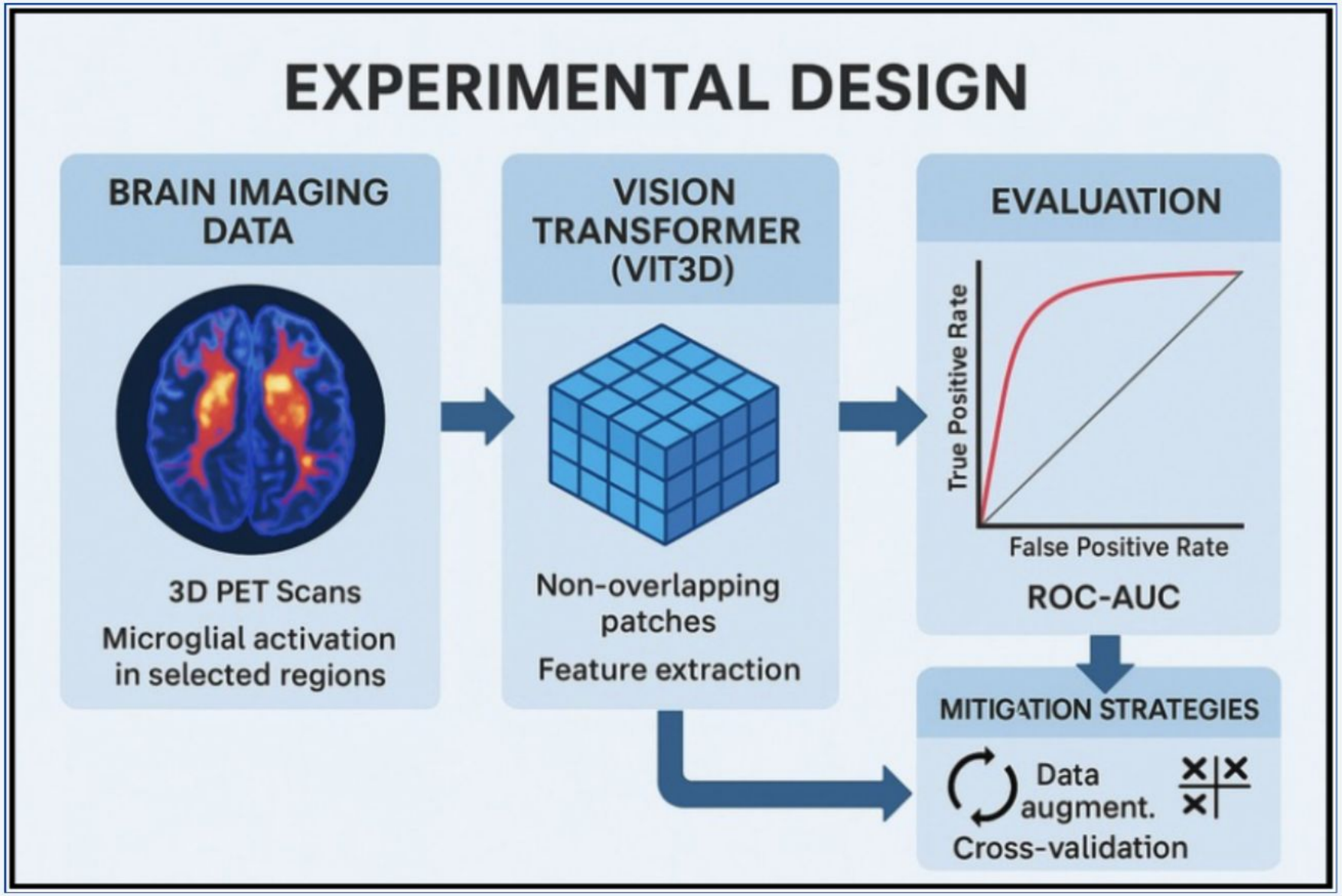


IMPACT

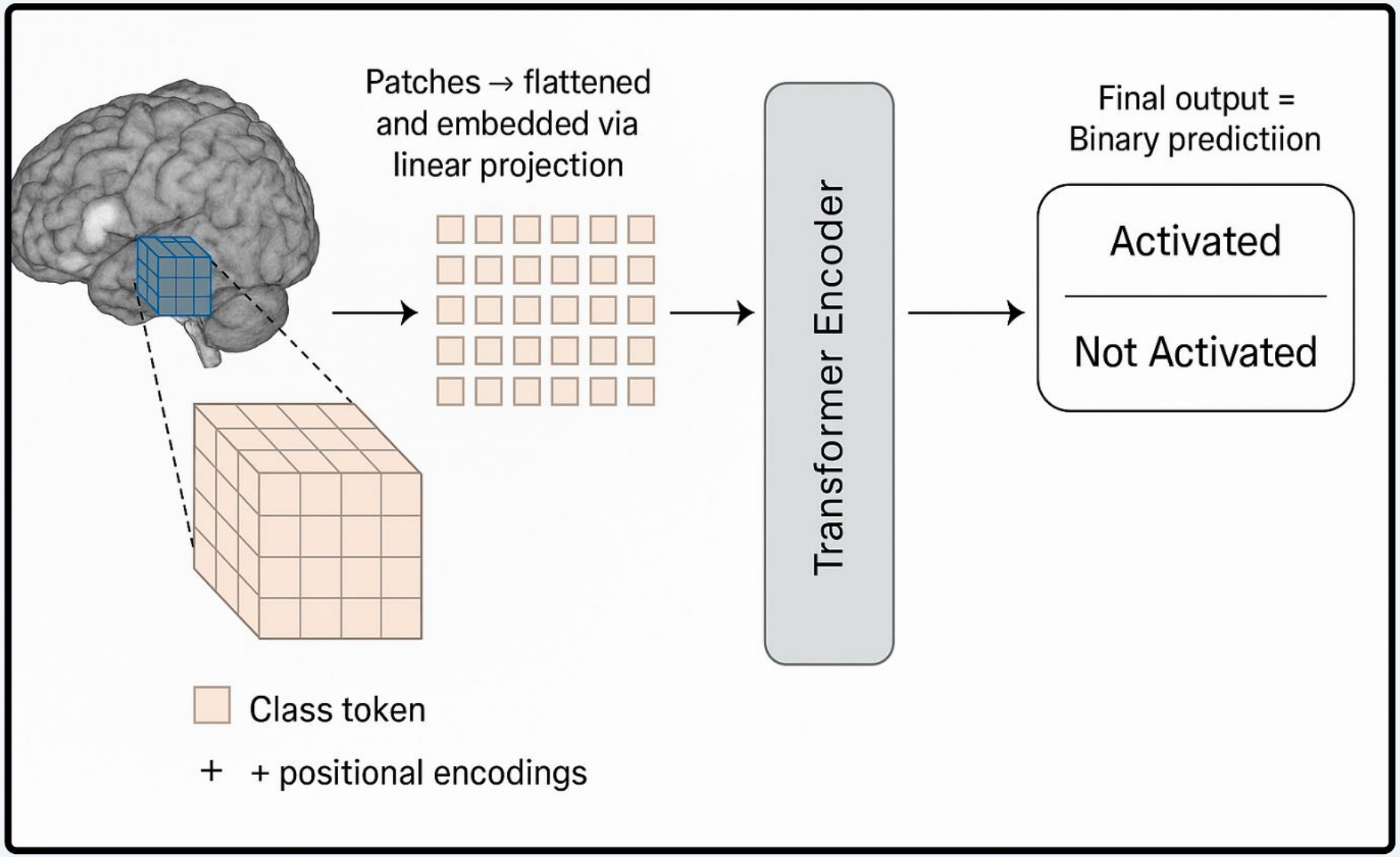
Transition to real data; improve patient outcomes



DEVICE DESIGN/METHODOLOGY

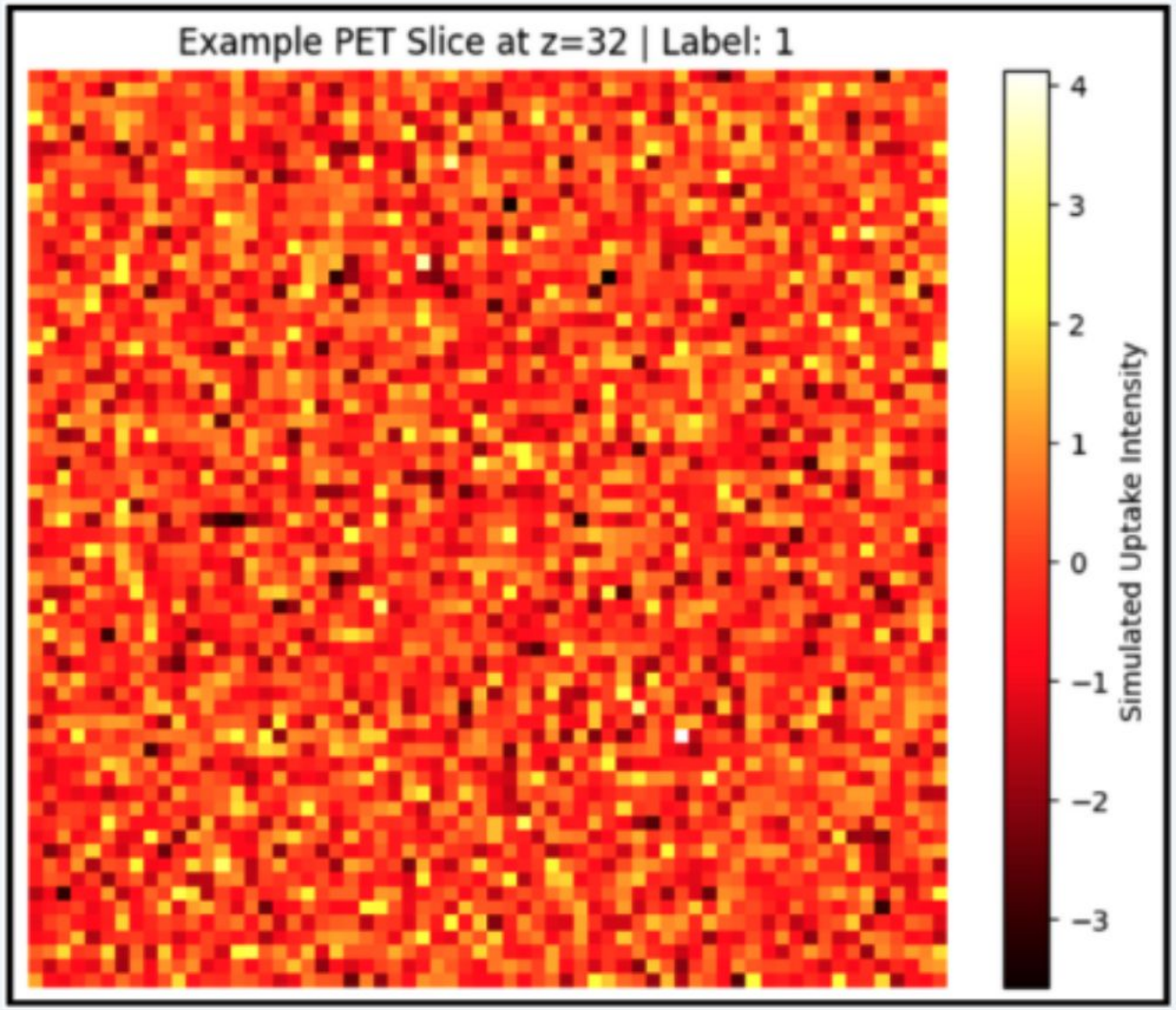


DEVICE DESIGN/METHODOLOGY



DEVICE DESIGN/METHODOLOGY

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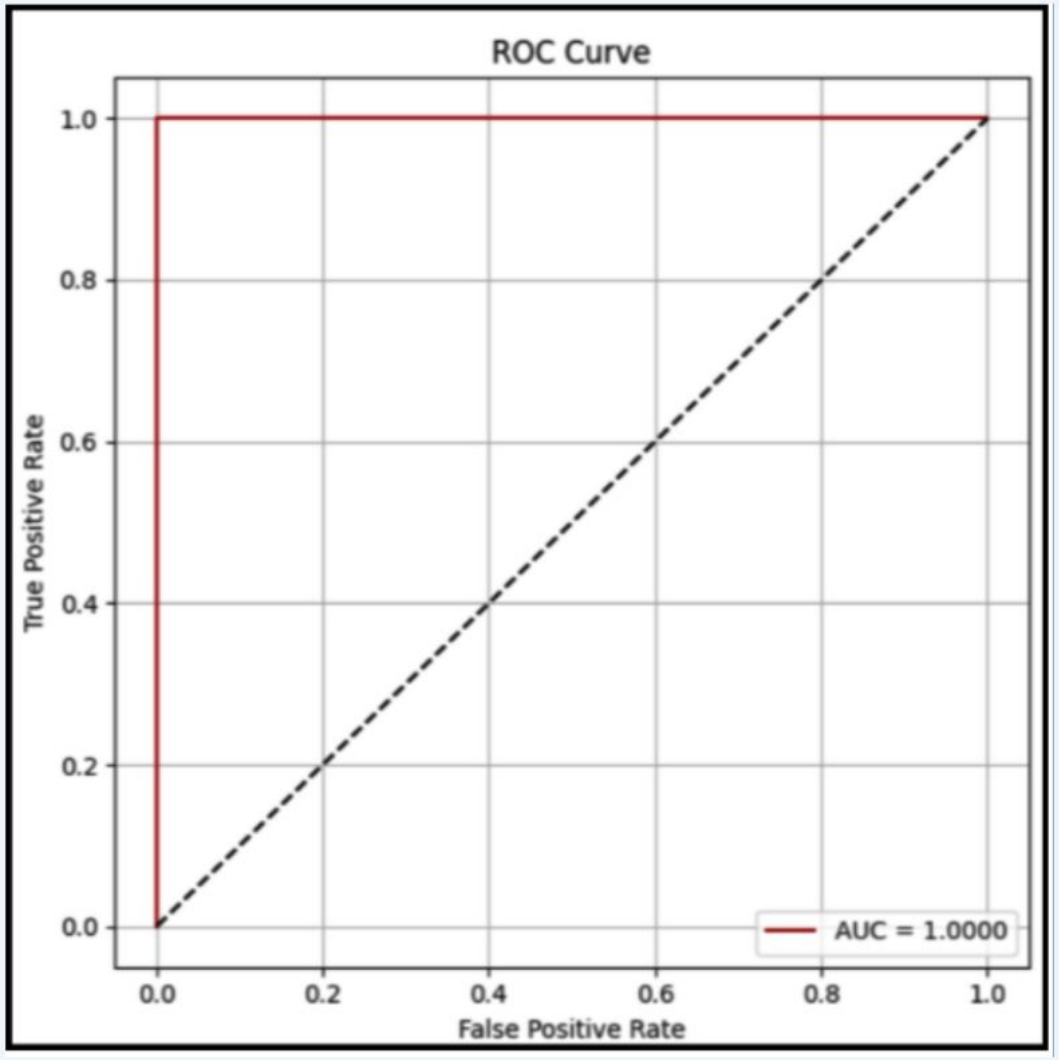


HOTSPOTS

UNIFORM NEGATIVES

RESULTS:

SYNTHETIC DATA TRAINING

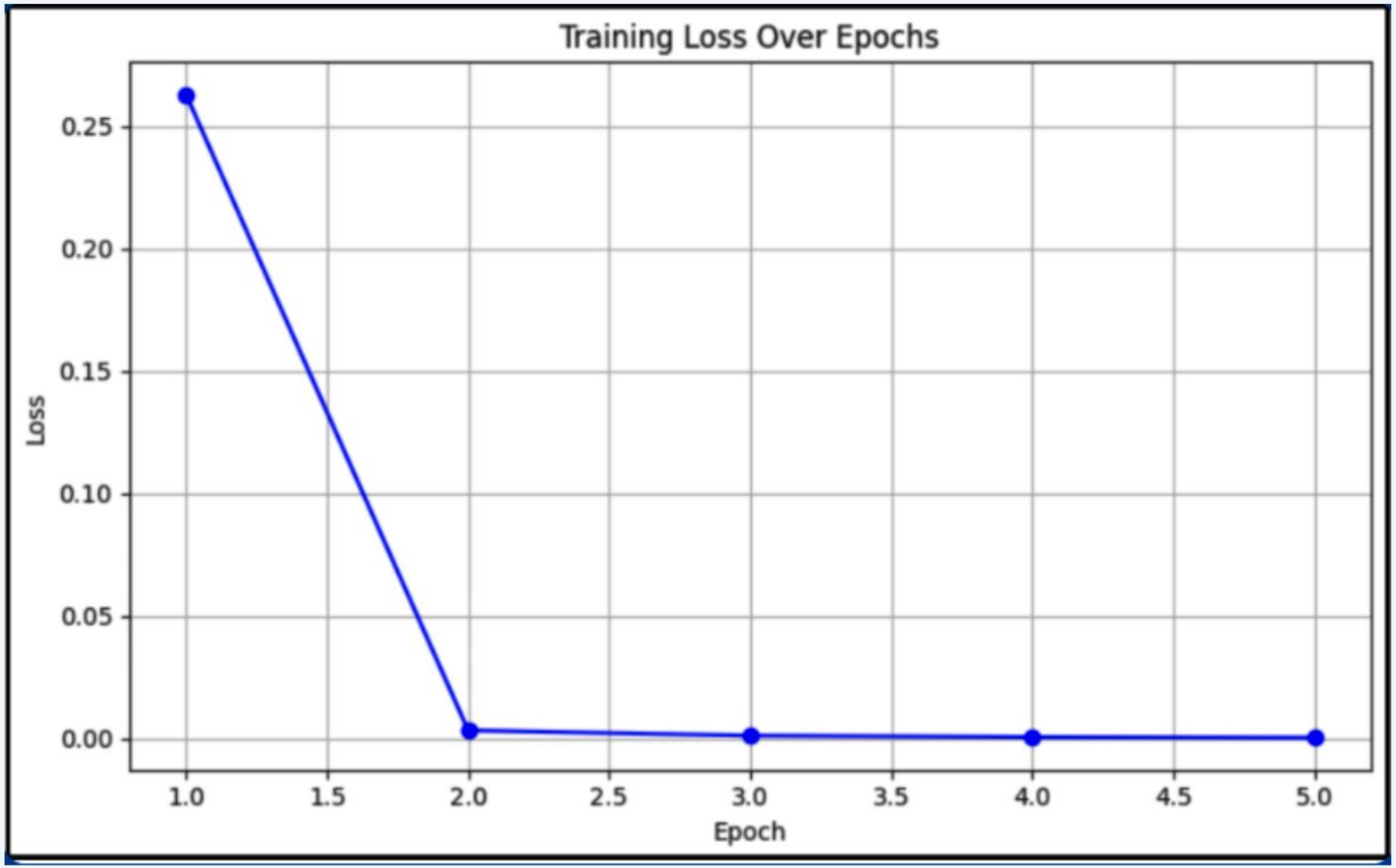


Near Perfect ROC-AUC Score across multiple trials

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RESULTS:

SYNTHETIC DATA TRAINING

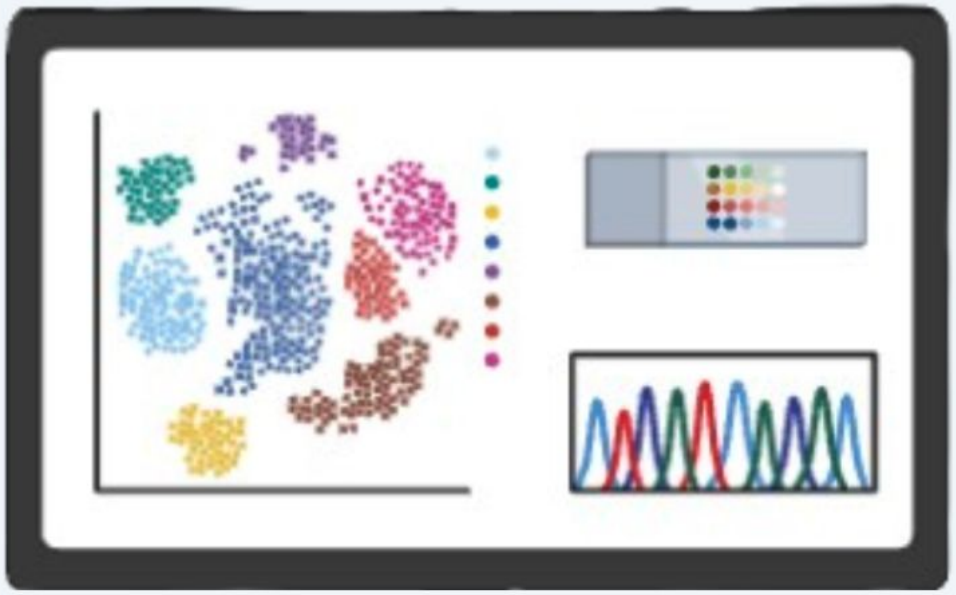


Rapid convergence within 5 epochs

CONCLUSIONS AND DISCUSSIONS

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FROM PROOF-OF-CONCEPT TO REAL IMPACT

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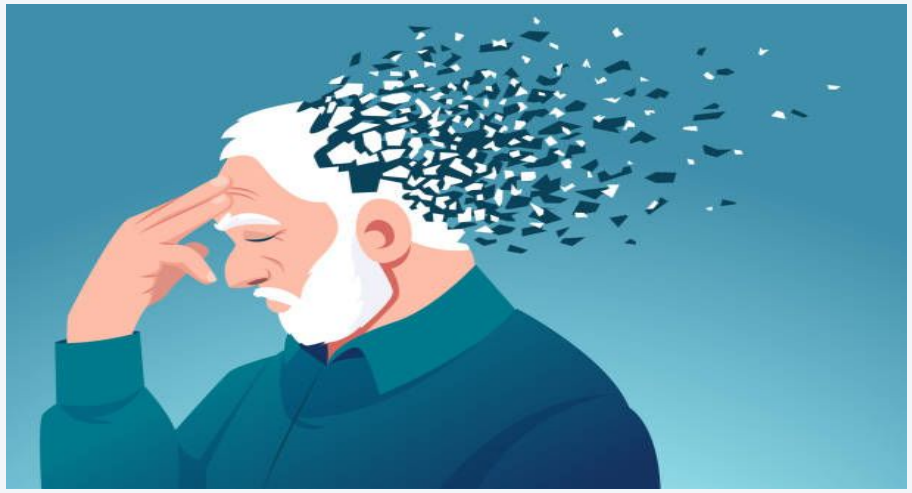
V Conclusion & Discussion

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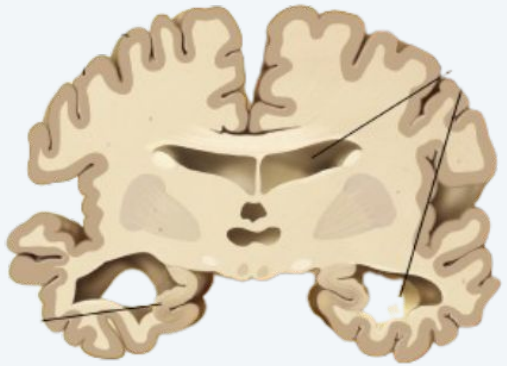








Healthy Brain



Severe AD

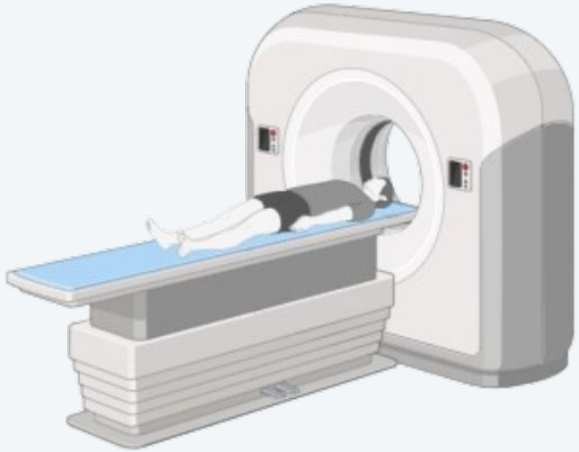
II Objectives/Research Focus

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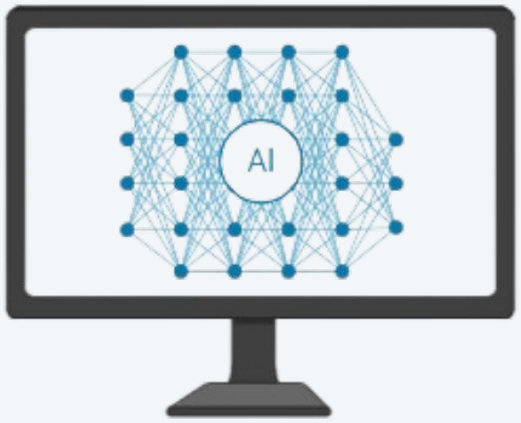
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Process

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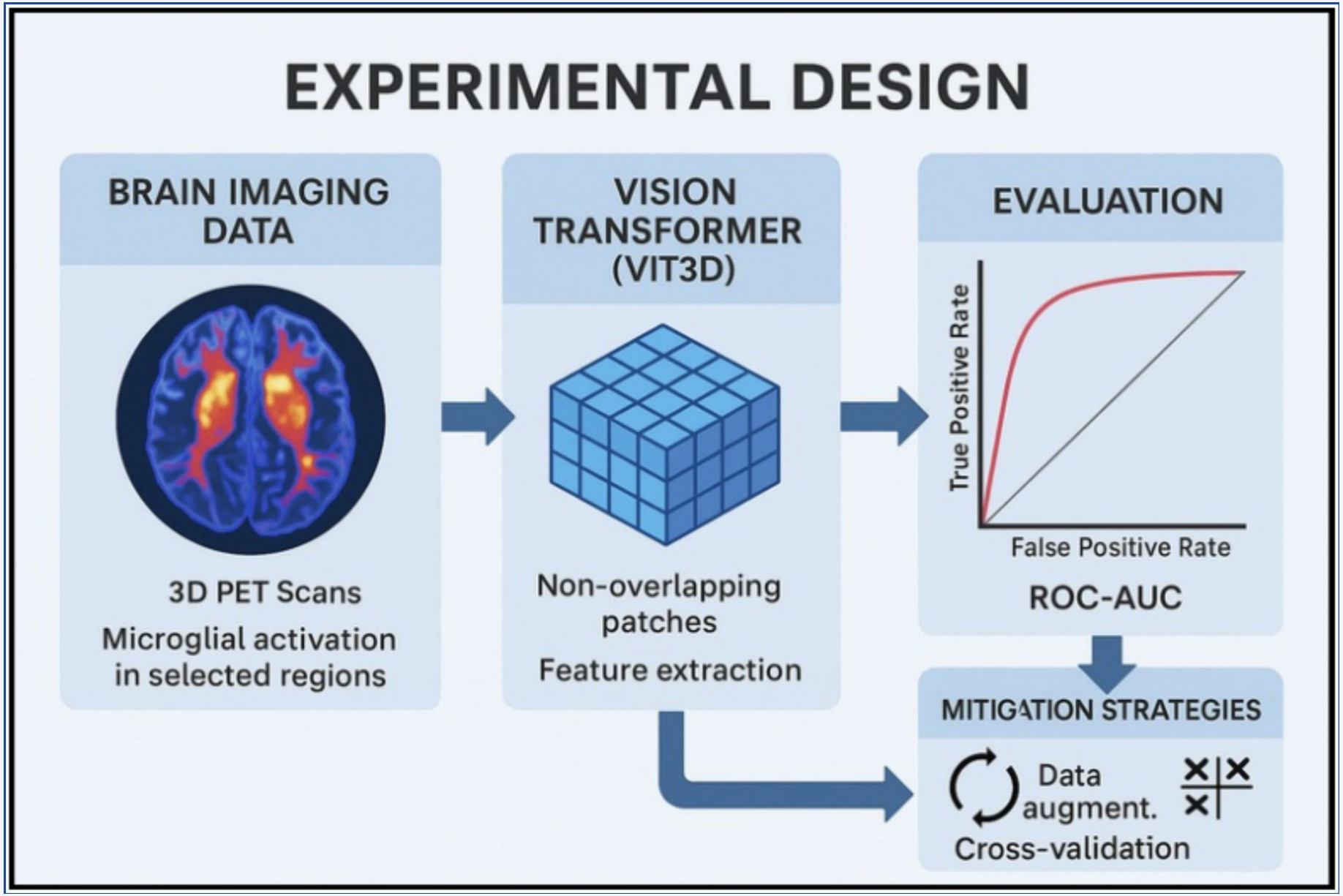


Impact

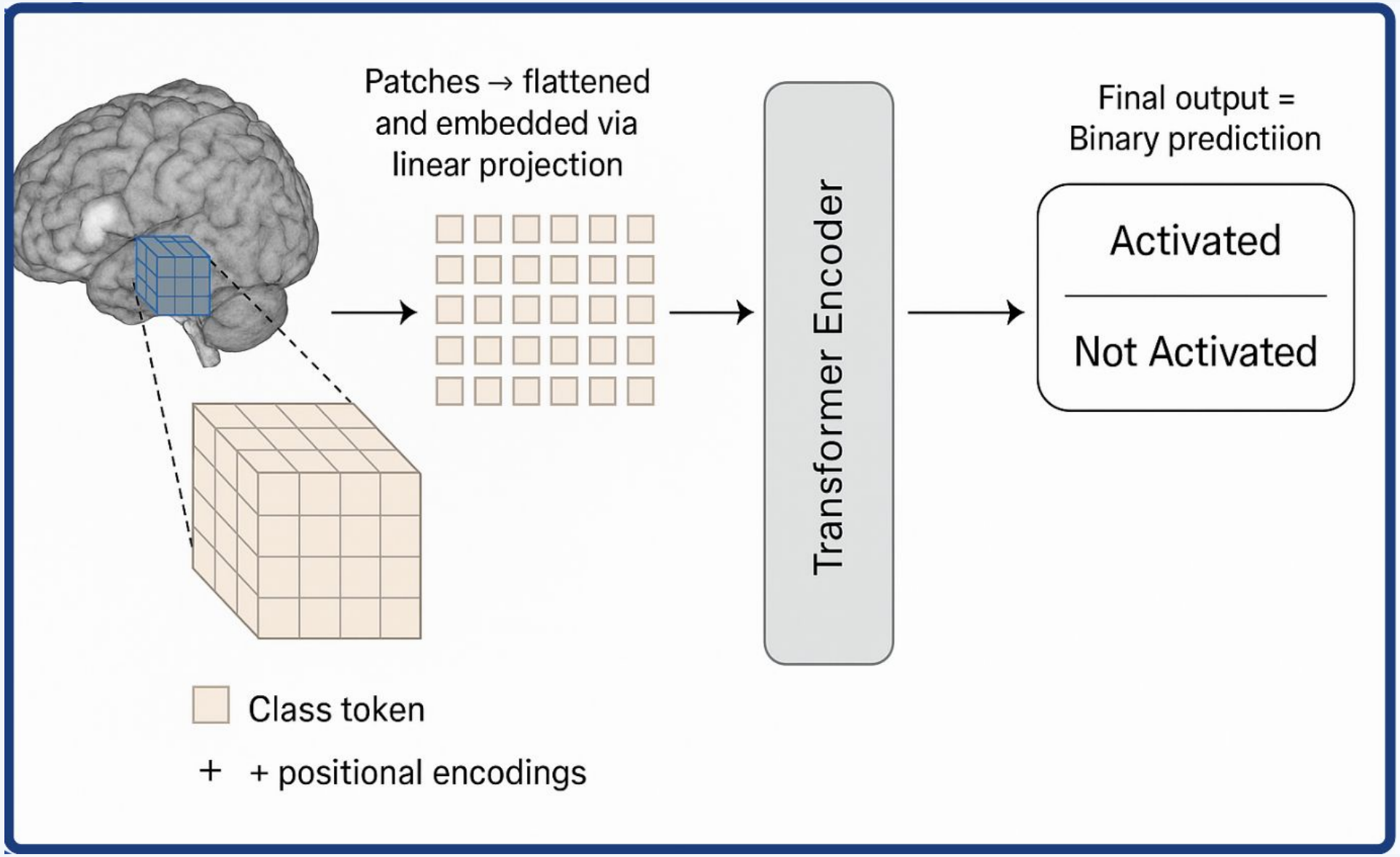
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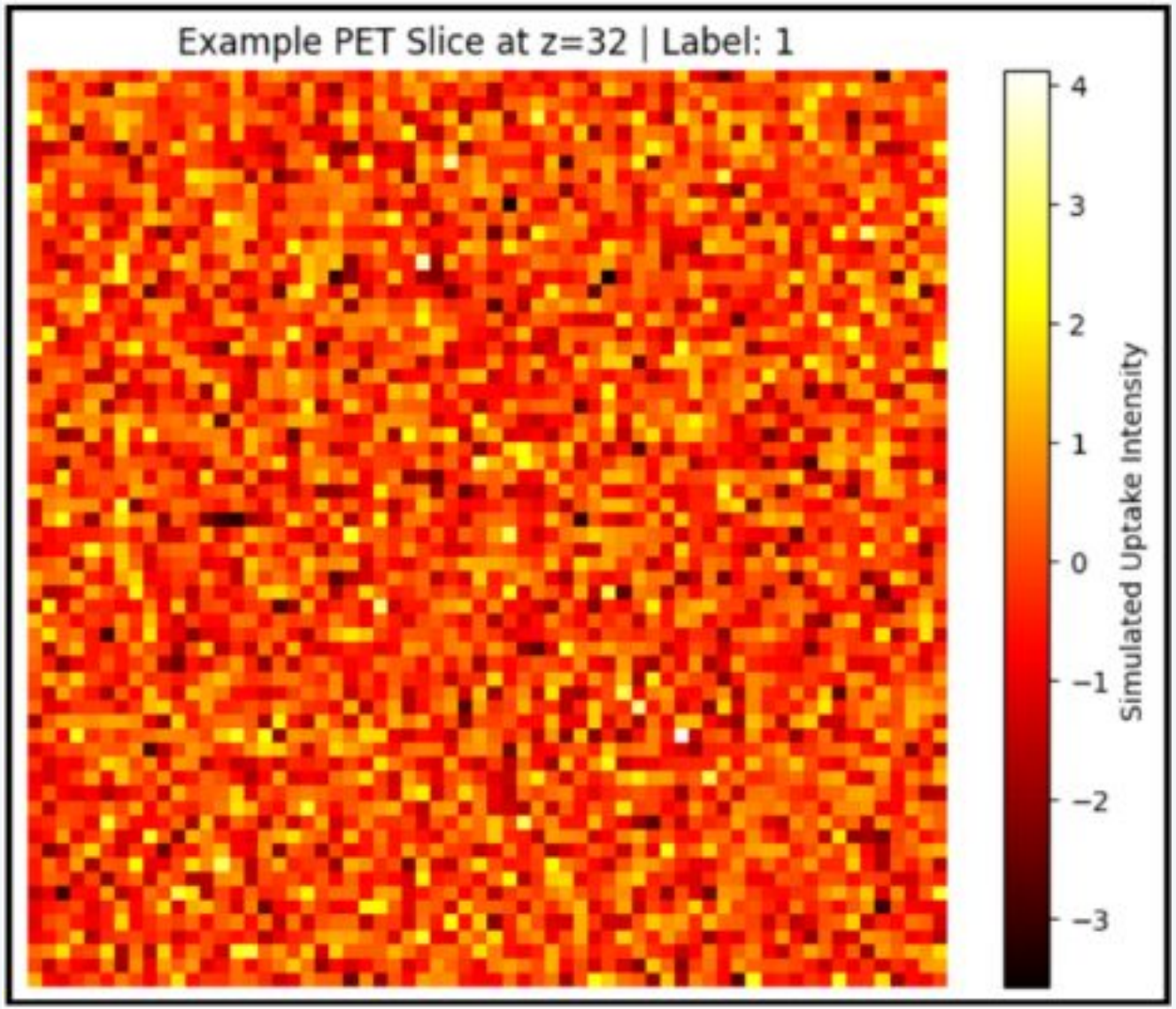
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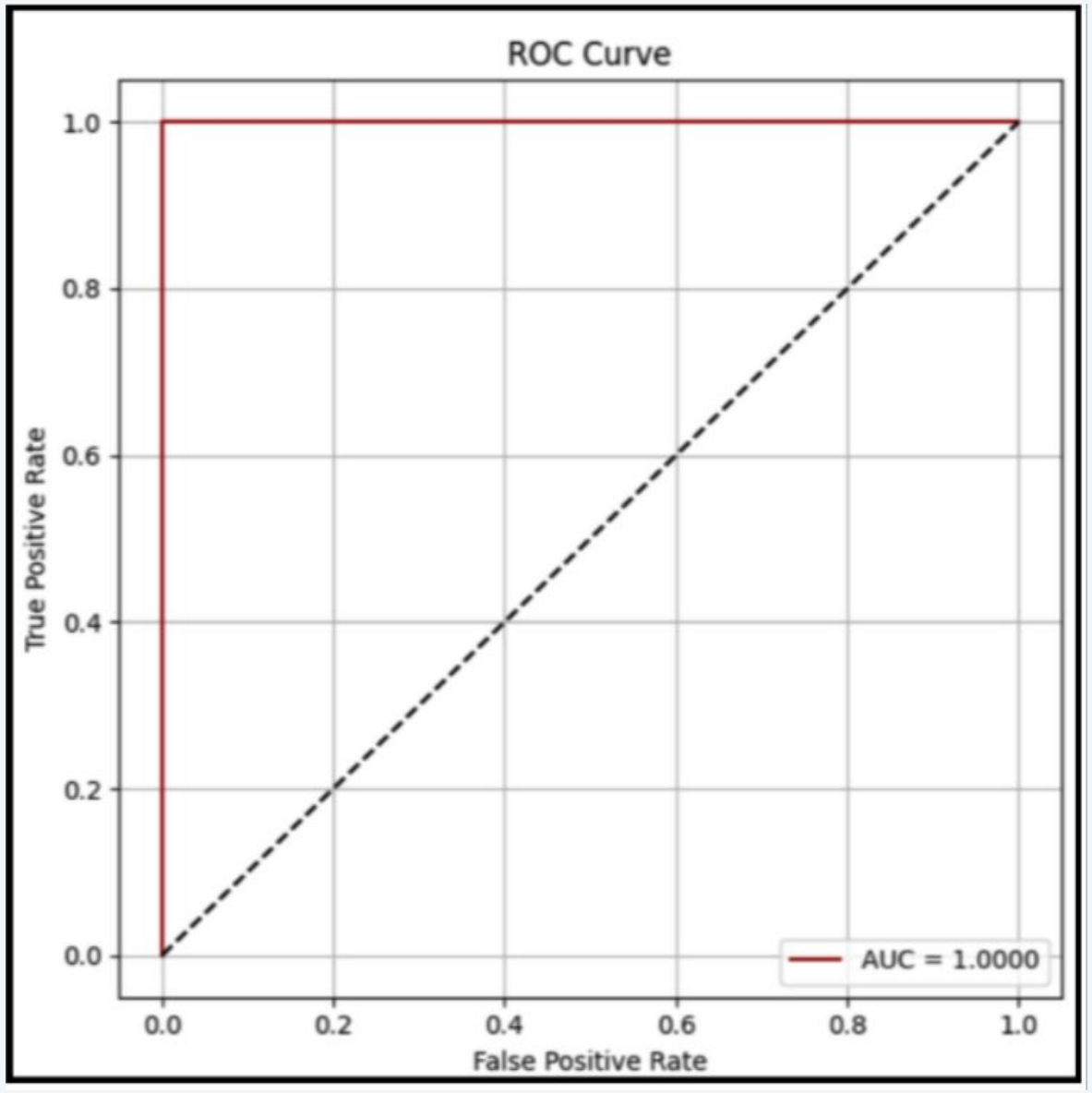
III Device Design/Methodology



Hotspots

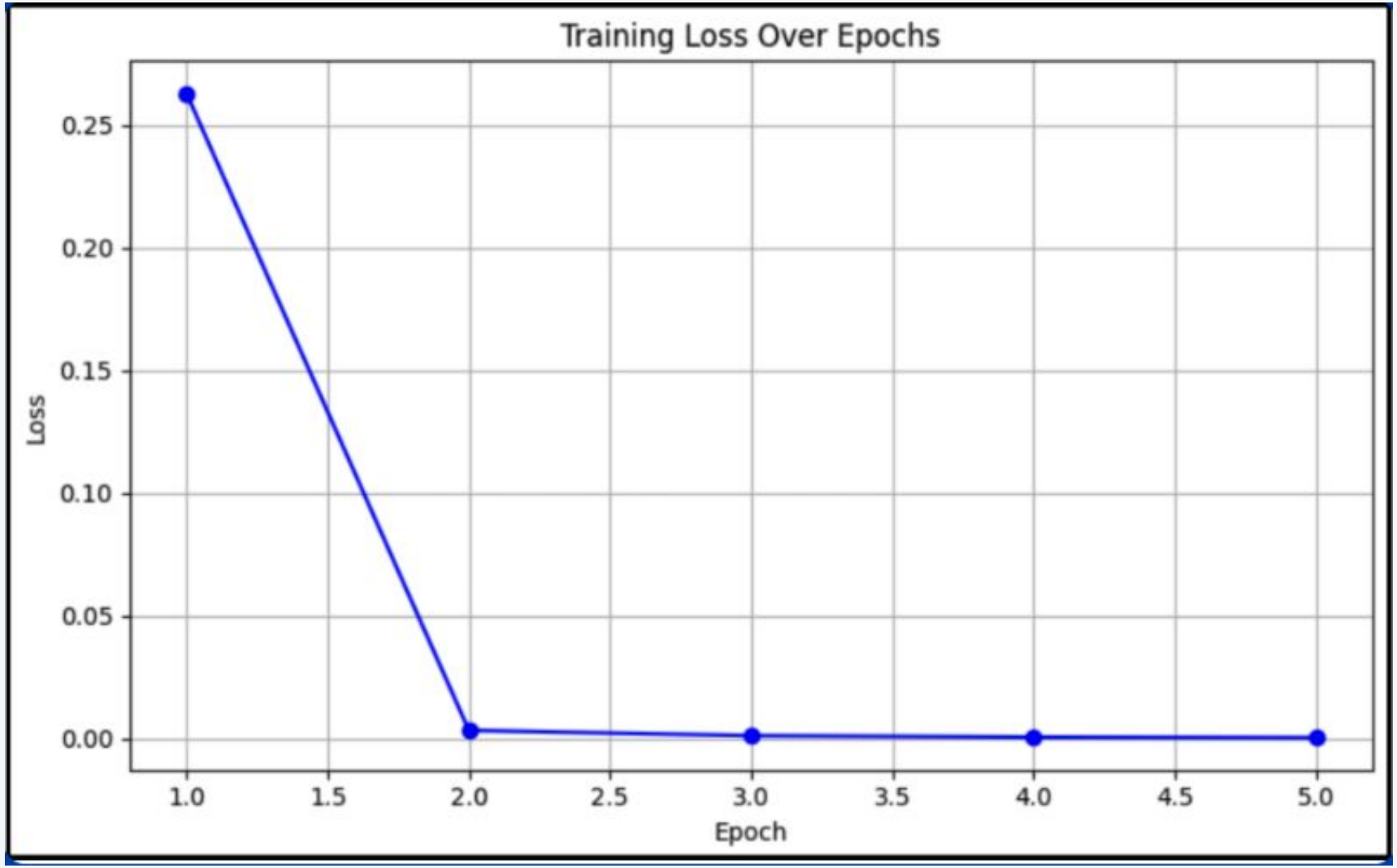
Uniform negatives

IV Results: Synthetic Data Training



Near Perfect ROC-AUC Score across multiple trials

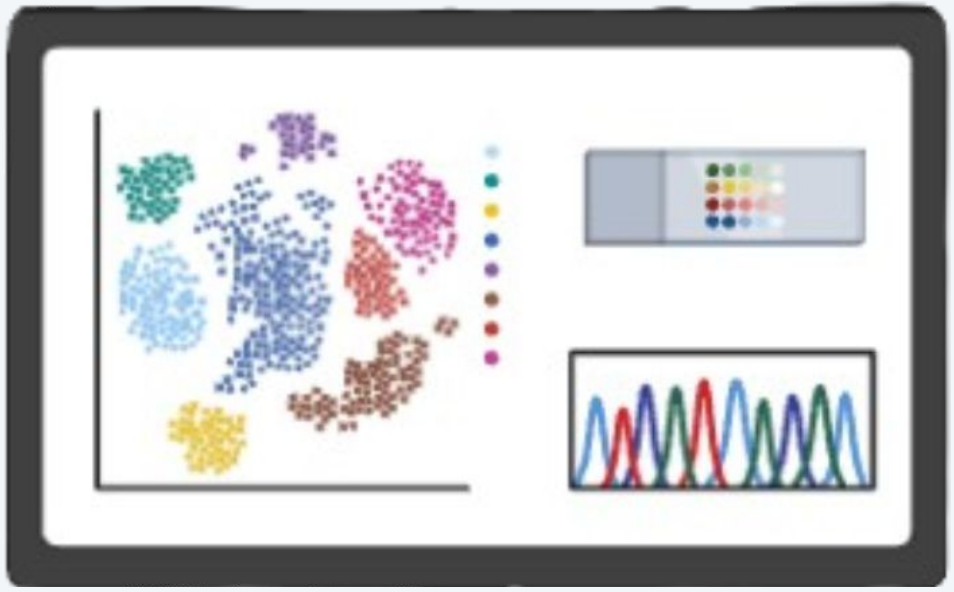
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V Conclusions and Discussions







From proof-of-concept to real impact

References

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